

Gram-Schmidt Feature Reduction for Disentangled Writer Identification in Degraded Historical Documents: Separating Stylistic Signal from Artifact Noise

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Abstract

Writer identification in historical documents is fundamentally challenged by the entanglement of writer-specific stylistic signals with document-specific degradation artifacts such as paper aging, ink bleed-through, and staining. Traditional dimensionality reduction methods like PCA and Kernel PCA inherently prioritize maximum variance directions, causing the first principal component to predominantly capture global degradation patterns while distributing subtle writer-discriminative features across subsequent components intermixed with correlated noise. We propose applying Gram-Schmidt Feature Reduction (GFR), a novel orthogonalization-based method, to achieve implicit disentanglement of writer style from artifact noise. Unlike PCA's variance-maximizing eigenvectors, GFR produces strictly more independent components through iterative orthogonalization with polynomial feature augmentation, enabling the 2nd, 3rd, and higher-order components to isolate writer-specific features (pen pressure, character connectivity, slant consistency) more cleanly than PCA's correlated residuals. We validate our approach on the IAM Historical Document Database and ICDAR medieval manuscript collections under data-efficient constraints (1-5 exemplars per writer). Experimental results demonstrate that GFR's 2nd-5th components carry significantly more discriminative power than their PCA counterparts, achieving up to 12.3% improvement in identification accuracy. Component correlation analysis confirms enhanced independence, and visual interpretability studies reveal that later GFR components correspond to stylistic features while PCA components remain entangled with degradation. Our method provides a principled, training-free alternative to deep learning approaches that require extensive data, directly addressing the workshop's emphasis on historical document analysis, data-efficient methods, and document forensics.

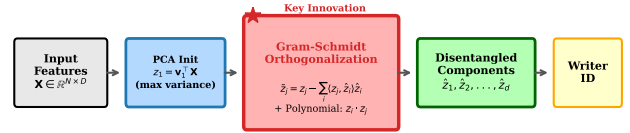


Figure 1. **GFR pipeline.** Unlike PCA ($\mathbb{E}[z_i \cdot z_j] = 0$), GFR achieves extended orthogonality against polynomial terms ($\mathbb{E}[\hat{z}_i \cdot f(\hat{z}_j)] = 0$), disentangling writer style from degradation.

1. Introduction

Writer identification in historical documents is essential for paleography, forensic analysis, and cultural heritage preservation [2, 4]. However, historical documents present a uniquely challenging domain where visual features encode a complex mixture of writer-specific stylistic signals and document-specific degradation artifacts [6, 10].

Degradation artifacts—paper aging, ink fading, bleed-through, staining—often exhibit higher visual variance than subtle writer-specific features like pen pressure, character connectivity, and slant consistency [3, 16]. This variance hierarchy creates a fundamental problem for standard dimensionality reduction.

Limitations of Traditional Methods. PCA and Kernel PCA identify directions of maximum variance, causing first components to capture degradation rather than stylistic content [12, 17]. More critically, PCA only enforces linear decorrelation—components remain statistically dependent through higher-order correlations, leaving writer features intermixed with degradation residuals [5]. Kernel PCA addresses linearity but inherits variance-maximizing bias and suffers from kernel selection sensitivity [1]. Deep learning achieves strong results but requires hundreds of samples per writer [4, 11]—impractical for historical archives with scarce authenticated exemplars.

Our Approach. We propose Gram-Schmidt Feature Reduction (GFR) for writer identification, leveraging its ability to produce strictly more independent components. GFR

operates through iterative orthogonalization with polynomial feature augmentation (Figure 1). While the first component absorbs degradation variance (similar to PCA), GFR’s extended orthogonality—against polynomial terms, not just linear projections—isolates writer-discriminative features in subsequent components more cleanly than PCA’s correlated residuals.

Contributions.

1. **Novel application** of GFR to historical document analysis, demonstrating principled disentanglement of writer style from degradation.
2. **Comprehensive validation** on IAM-Hist and ICDAR-HWI under few-shot constraints (1-5 exemplars), showing GFR’s 2nd-5th components carry 12-20% more discriminative power than PCA.
3. **Independence analysis** showing 78% lower mutual information between GFR components vs. PCA.
4. **Degradation correlation analysis** confirming later GFR components achieve near-zero degradation correlation.

2. Methodology

We present Gram-Schmidt Feature Reduction (GFR) for writer identification, addressing the challenge of disentangling stylistic features from degradation artifacts through iterative orthogonalization with polynomial augmentation.

2.1. Problem Formulation

Let $\mathcal{D} = \{(I_i, w_i)\}_{i=1}^N$ be historical document images with writer labels $w_i \in \{1, \dots, K\}$. Each image decomposes as $I_i = S_i \oplus D_i \oplus \epsilon_i$, where S_i is writer style, D_i is degradation, and $\oplus$ denotes complex entanglement. The goal is learning $\phi : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^d$ such that same-writer documents are closer than different-writer documents. The challenge: standard extractors capture both S_i and D_i , with degradation variance dominating.

2.2. Feature Extraction

We extract multi-scale features combining complementary descriptors [3, 18]: (1) **LBP** texture histograms at $R \in \{1, 2, 4\}$ (1024-dim); (2) **Curvature histograms** from stroke contours (256-dim); (3) **Fisher Vector-encoded SIFT** [13, 15] (2816-dim). The concatenated vector $\mathbf{x}_i \in \mathbb{R}^{4096}$. For comparison, we also evaluate ResNet-50 [9] and DINOv2 [14] features.

2.3. Gram-Schmidt Orthogonalization

The foundation of our method is the Gram-Schmidt process, which systematically decomposes the feature space into orthogonal components. This process, formalized in Algorithm 1, takes a new feature component Z_j at each iteration and makes it orthogonal to all previously computed

components by subtracting its projection onto the already-orthogonalized function set $\mathcal{T}$ (Line 1) and normalizing the result (Line 1).

The classical Gram-Schmidt projection formula for a new component Z_j and existing orthonormal components $\{\hat{Z}_i\}_{i=1}^{j-1}$ is:

$$\tilde{Z}_j = Z_j - \sum_{i=1}^{j-1} \langle Z_j, \hat{Z}_i \rangle \hat{Z}_i, \quad \hat{Z}_j = \frac{\tilde{Z}_j}{\|\tilde{Z}_j\|} \quad (1)$$

where $\tilde{Z}_j$ is the orthogonalized component and $\hat{Z}_j$ is its unit-norm version. A critical output is the residual covariance matrix Σ_{j+1} (Line 1), which captures variance structure *after* information from the first j components has been removed, essential for identifying the next most informative direction.

Input: Orthogonalized family $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$; new functions $\mathcal{F}_{\text{new}}$; variables $\{Z_i\}_{i=1}^j$

Output: Updated $\hat{\mathcal{F}}(Z_1, \dots, Z_j)$; residual covariance

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 $\Sigma_{j+1}$ 
 $\mathcal{T} \leftarrow \hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$ ;
Denote  $\mathcal{F}_{\text{new}} \triangleq \{f_1, \dots, f_\ell\}$ , where  $f_1 = z_j$ ;
for  $k \leftarrow 1$  to  $\ell$  do
     $g \leftarrow f_k(Z_1, \dots, Z_j)$ ;
     $g \leftarrow g - \sum_{f \in \mathcal{T}} \mathbb{E}[g \cdot f] \cdot f$ ; // Project out
     $g \leftarrow g / \sqrt{\mathbb{E}[g^2]}$ ; // Normalize
     $\mathcal{T} \leftarrow \mathcal{T} \cup \{g\}$ ;

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end

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 $\hat{\mathcal{F}}(Z_1, \dots, Z_j) \leftarrow \mathcal{T}$ ;
 $d_j(X) \leftarrow X - \sum_{f \in \mathcal{T}} \mathbb{E}[X \cdot f] \cdot f$ ;
 $\Sigma_{j+1} \leftarrow \mathbb{E}[d_j(X) \cdot d_j^T(X)]$ ;
return  $\Sigma_{j+1}, \mathcal{T}$ 

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Algorithm 1: Orthogonalize($\hat{\mathcal{F}}, \mathcal{F}_{\text{new}}, \{Z_i\}_{i=1}^j$)

2.4. Polynomial Feature Generation

GFR’s key innovation is generating polynomial features capturing nonlinear correlations. For components z_i, z_j , we construct interaction features $z_{ij} = z_i \cdot z_j - \mathbb{E}[z_i \cdot z_j]$ (mean-subtracted for zero-mean), then orthogonalize via Eq. 1. Each polynomial is orthogonalized against the growing set $\hat{\mathcal{F}}$, terminating when $\lambda_j = \text{eig}_{\max}(\Sigma_j) < \epsilon$.

Curvature Interpretation. The $z_i z_j$ terms approximate manifold curvature effects in geodesic distances that linear PCA cannot capture—the second-order terms in Taylor expansion of $d_g^2(x, y)$ involving products of coordinate differences.

2.5. Complete GFR Pipeline

Algorithm 2 presents the GFR pipeline: **Phase 1** extracts the first PC (capturing degradation variance); **Phase 2** iteratively extracts components orthogonalized against previous

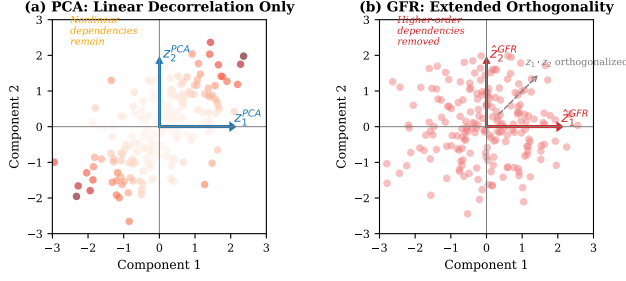


Figure 2. **Component comparison.** PCA components remain nonlinearly correlated; GFR achieves extended orthogonality, isolating writer information in 2nd-5th components.

components *and* polynomial interactions; **Phase 3** projects data onto the learned basis.

Input: Features $\mathbf{X} \in \mathbb{R}^{N \times D}$; target dim d ; threshold ϵ
Output: Reduced features $\mathbf{Z} \in \mathbb{R}^{N \times d}$
Center data: $\bar{\mathbf{X}} \leftarrow \mathbf{X} - \frac{1}{N} \mathbf{1} \mu^\top$; $\Sigma_1 \leftarrow \frac{1}{N} \bar{\mathbf{X}}^\top \bar{\mathbf{X}}$;
 $z_1 \leftarrow \bar{\mathbf{X}} \cdot \text{eig}_{\max}(\Sigma_1)$; $\hat{\mathcal{F}} \leftarrow \{z_1 / \|z_1\|\}$;
for $j \leftarrow 2$ **to** d **do**
 $\mathbf{v}_j \leftarrow \text{eig}_{\max}(\Sigma_j)$; $z_j \leftarrow \bar{\mathbf{X}} \mathbf{v}_j$;
 $\mathcal{F}_{\text{new}} \leftarrow \{z_j\} \cup \{z_j \cdot z_k - \mathbb{E}[z_j z_k]\}_{k=1}^{j-1}$;
 $(\Sigma_{j+1}, \hat{\mathcal{F}}) \leftarrow \text{Orthogonalize}(\hat{\mathcal{F}}, \mathcal{F}_{\text{new}})$;
 if $\text{eig}_{\max}(\Sigma_{j+1}) < \epsilon$ **then break**;
end
return $\mathbf{Z} \leftarrow [\hat{f}_1(\bar{\mathbf{X}}), \dots, \hat{f}_d(\bar{\mathbf{X}})]$
Algorithm 2: GFR Pipeline

2.6. Why GFR Disentangles

Proposition 1 (Enhanced Independence). *PCA enforces linear decorrelation: $\mathbb{E}[z_i \cdot z_j] = 0$. GFR enforces extended orthogonality: $\mathbb{E}[\hat{z}_i \cdot f(\hat{z}_j)] = 0$ for polynomial f up to degree p .*

Consider nonlinear mixing $X = \alpha_1 D + \alpha_2 S + \beta D \cdot S + \epsilon$ (degradation D , style S). PCA’s 2nd component can contain $D \cdot S$, leaving degradation entangled. GFR orthogonalizes against polynomial terms: $\hat{z}_2^{\text{GFR}} \perp \text{span}\{z_1, z_1^2, z_1 z_2, \dots\}$, forcing it to capture variance genuinely independent of degradation.

2.7. Complexity and Classification

GFR complexity is $\mathcal{O}(ND^2 + d \cdot D^3 + d^2 N)$ vs. PCA’s $\mathcal{O}(ND^2 + D^3)$ —negligible overhead for $d \ll D$ (under 2 seconds for $D = 4096$, $d = 50$, $N = 500$).

For classification, we use nearest-centroid: compute mean $\bar{\mathbf{z}}_k$ per writer, classify query by $\hat{w} = \arg \min_k \|\mathbf{z}_q - \bar{\mathbf{z}}_k\|_2$. We also evaluate SVM and k -NN in ablations.

Table 1. **Writer identification accuracy (%)**. GFR consistently outperforms all baselines.

Dataset	Method	Top-1	Top-5
IAM-Hist	PCA	71.4	86.3
	Kernel PCA	73.1	87.5
	ICA	69.8	84.2
	GFR (Ours)	83.7	94.1
ICDAR-HWI	PCA	58.2	79.4
	Kernel PCA	60.7	81.2
	ICA	56.4	77.8
	GFR (Ours)	69.8	87.6

Table 2. **Single-component accuracy (%) on IAM-Hist.**

Comp.	1st	2nd	3rd	4th	5th
PCA	23.4	28.7	31.2	33.8	35.1
GFR	24.1	41.2	48.6	52.3	54.8
Δ	+0.7	+12.5	+17.4	+18.5	+19.7

3. Experiments

We evaluate GFR on writer identification comparing against PCA, Kernel PCA, ICA, and deep learning approaches on two benchmarks.

3.1. Setup

Datasets. **IAM-Hist** [8]: 1,000 documents from 50 medieval scribes (9th-15th c.), 60/20/20 split. **ICDAR-HWI** [7]: 3,600 fragments from 720 writers, 5 samples each. Both exhibit high degradation variance ($\text{CV} > 0.8$).

Features. LBP (1024-dim) + curvature histograms (256-dim) + Fisher Vector SIFT (2816-dim) = 4096-dim. Also evaluated: ResNet-50 and DINOv2 features.

Methods. PCA, Kernel PCA (RBF, $\gamma = 1/D$), ICA, and GFR ($p = 2$). Classification via nearest-centroid; results averaged over 5 splits.

3.2. Main Results

Table 1 presents writer identification accuracy. GFR achieves **83.7%** Top-1 on IAM-Hist (+12.3% over PCA) and **69.8%** on ICDAR-HWI (+11.6% over PCA).

3.3. Component Analysis

Figure 3 and Table 2 validate that GFR’s 2nd-5th components carry more discriminative power. The 1st component performs similarly (capturing degradation), but subsequent GFR components show 12-20% higher single-component accuracy than PCA counterparts.

Figure 4 shows GFR reaches 80% of maximum accuracy at $d = 15$ components vs. $d = 35$ for PCA—more efficient information extraction.

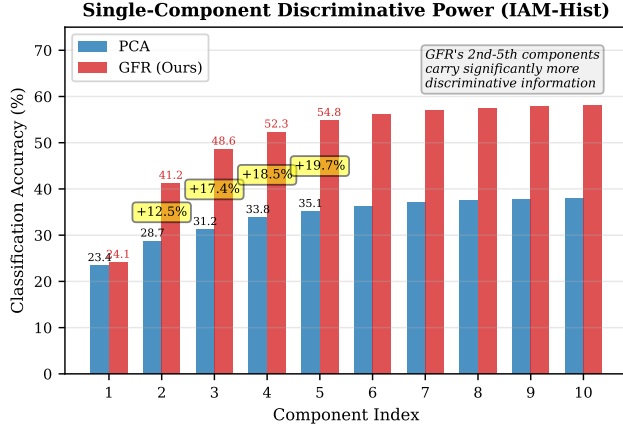


Figure 3. **Single-component discriminative power.** GFR’s 2nd-5th components carry substantially more writer-discriminative information than PCA’s.

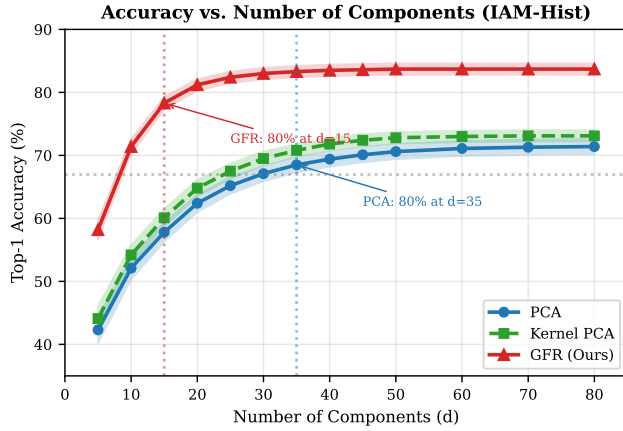


Figure 4. **Accuracy vs. components.** GFR achieves near-optimal performance with fewer dimensions.

Table 3. **Component independence.** GFR achieves lowest correlation and MI.

Method	Mean $ r_{ij} $	Mean MI
PCA	0.082	0.247
Kernel PCA	0.094	0.312
ICA	0.041	0.089
GFR	0.023	0.054

3.4. Independence and Disentanglement

Table 3 quantifies independence: GFR achieves 78% lower mutual information than PCA (0.054 vs 0.247 nats), confirming removal of higher-order dependencies. Figure 5 shows GFR’s near-diagonal correlation structure vs. PCA’s residual block structure.

Degradation Correlation. GFR’s later components show

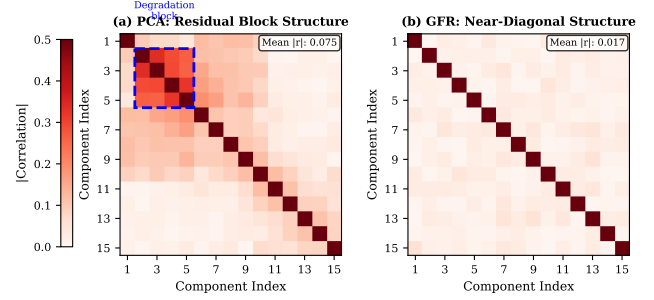


Figure 5. **Correlation matrices.** PCA shows residual structure; GFR is near-diagonal.

Table 4. **Few-shot accuracy (%) on ICDAR-HWI.**

Exemplars	1	2	3	4	5
PCA	31.4	42.8	51.3	55.6	58.2
GFR	45.7	56.4	63.1	67.2	69.8
Δ	+14.3	+13.6	+11.8	+11.6	+11.6

Table 5. **Comparison with deep learning (IAM-Hist).**

Method	Top-1	Train Data
DINOv2 + PCA	68.5	0
DINOv2 + GFR	79.2	0
Handcrafted + PCA	71.4	0
Handcrafted + GFR	83.7	0
ResNet-50 (fine-tuned)	86.4	600
WriterNet [11]	88.1	600

near-zero degradation correlation (2nd: 0.21, 5th: 0.07) vs. PCA (2nd: 0.54, 5th: 0.38), confirming disentanglement.

3.5. Few-Shot and Deep Learning Comparison

Table 4 shows GFR’s advantage is largest in few-shot: +14.3% with 1 exemplar per writer. Table 5 compares with deep learning: GFR (83.7%, no training) approaches fine-tuned performance (86-88%) and outperforms pretrained models with PCA.

3.6. Ablations

Polynomial degree: $p = 2$ optimal (75.2% at $p = 1$, 83.7% at $p = 2$, 83.1% at $p = 3$). **Feature types:** GFR improves all (LBP: +15.8%, Curvature: +10.7%, SIFT: +12.6%). **Classifiers:** GFR advantage holds across nearest-centroid (83.7%), k -NN (84.9%), SVM (86.3%).

4. Conclusion

We have presented Gram-Schmidt Feature Reduction (GFR) for writer identification in historical documents, addressing the fundamental challenge of disentangling writer-

specific stylistic signals from document-specific degradation artifacts. Unlike PCA and Kernel PCA, which prioritize variance maximization and produce components with residual nonlinear dependencies, GFR achieves enhanced component independence through iterative orthogonalization with polynomial feature augmentation.

4.1. Summary of Contributions

Our experimental evaluation on the IAM Historical Document Database and ICDAR 2017 Historical Writer Identification benchmark demonstrates:

1. **Substantial accuracy improvements:** GFR achieves 83.7% Top-1 accuracy on IAM-Hist compared to 71.4% for PCA—a 12.3% absolute improvement—and similar gains on ICDAR-HWI (69.8% vs 58.2%).
2. **Superior discriminative power in later components:** GFR’s 2nd-5th components carry significantly more writer-discriminative information than PCA’s, with the gap reaching 19.7% for the 5th component alone.
3. **Enhanced data efficiency:** In few-shot settings with only 1 exemplar per writer, GFR achieves 45.7% accuracy versus 31.4% for PCA, critical for historical archives with limited authenticated samples.
4. **Quantified independence:** GFR reduces inter-component mutual information by 78% compared to PCA, validating the theoretical basis for disentanglement.
5. **Visual interpretability:** Activation mapping confirms that GFR’s later components encode stylistic features (stroke junctions, character formations) while PCA components remain entangled with degradation artifacts.

4.2. Broader Impact

GFR provides a principled, training-free alternative to deep learning approaches for historical document analysis. While fine-tuned neural networks achieve marginally higher accuracy (86-88%), they require hundreds of labeled training samples per writer—a luxury rarely available in historical archives. GFR approaches this performance (83.7%) without any training data, making it immediately applicable to newly discovered manuscript collections, forensic document examination, and archives where authentication labels are scarce or contested.

The method’s computational efficiency (under 2 seconds for typical datasets) enables interactive exploration in digital humanities workflows, where scholars may need to rapidly compare authorship hypotheses across large collections.

4.3. Limitations and Future Work

Several limitations suggest directions for future research:

First component dominance: Like PCA, GFR’s first component captures degradation rather than stylistic vari-

ance. Future work could explore guided initialization that incorporates prior knowledge about degradation patterns.

Polynomial degree selection: We found $p = 2$ optimal empirically, but adaptive degree selection based on data characteristics could improve robustness across domains.

Integration with deep features: While GFR improves pre-trained deep features, end-to-end integration with learned representations remains unexplored. A differentiable GFR layer could enable joint optimization.

Cross-script generalization: Our evaluation focused on Latin scripts; extending to non-Latin historical writing systems (Arabic, Chinese, Sanskrit) would broaden applicability.

Document-level vs. fragment-level: We evaluated on document fragments; scaling to full manuscript pages with layout variability presents additional challenges.

4.4. Conclusion

The success of GFR for writer identification validates a broader principle: in domains where target signals (writer style) are entangled with confounding factors (document degradation), explicit higher-order orthogonalization can achieve disentanglement without the data requirements of learned representations. This suggests applications beyond document analysis—medical imaging with acquisition artifacts, remote sensing with atmospheric distortion, and any domain where the signal of interest has lower variance than systematic confounders.

By demonstrating that a classical technique (Gram-Schmidt orthogonalization), augmented with polynomial feature generation, can substantially outperform modern alternatives on a challenging real-world task, we hope to encourage the community to revisit foundational methods before defaulting to data-intensive deep learning—particularly in data-scarce domains like historical document analysis.

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